



Reg. No.

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BCACACS 301

**Third Semester B.C.A. Degree Examination, December 2025/January 2026
(SEP) (2025-26 Batch Onwards)
DATABASE MANAGEMENT SYSTEM**

Time : 3 Hours

Max. Marks : 80

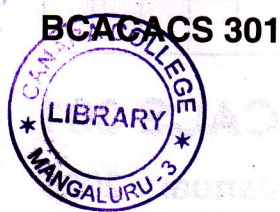
Note : Answer **any ten** questions from Part – A and **one full** question from **each** Unit in Part – B.

PART – A

(10×2=20)

1. a) What is database ? Define DBMS.
- b) Who is a DBA ? What are the responsibilities of a DBA ?
- c) Write any two advantages of DBMS.
- d) Define Tuple and Attribute.
- e) What is relational model ?
- f) What is dual table ?
- g) Write the syntax of delete command.
- h) How can you rename the table ? Give example.
- i) Give two advantages of PL/SQL over SQL.
- j) What is the purpose of COMMIT command ?
- k) What is a trigger ? What are three parts of trigger ?
- l) Mention any four implicit cursor attributes.

P.T.O.



PART – B

Unit – I

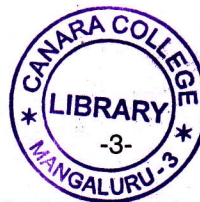
2. a) What are the characteristics of DBMS ? Explain.
b) Explain the classification of Database Management Systems.
c) Explain any five symbols used in an ER diagram with an example. (4+6+5)
3. a) Explain the three schema architecture of database with a neat diagram.
b) Explain different types of attributes with example.
c) What are the User friendly interfaces provided by DBMS ? Explain any four. (5+5+5)

Unit – II

4. a) Explain the characteristics of relations.
b) Explain 1NF with example.
c) Explain entity integrity and referential integrity constraints. (5+5+5)
5. a) Explain functional dependencies with example.
b) Explain different update operation in a Relational model.
c) Explain 2NF with an example. (5+5+5)

Unit – III

6. a) Explain CHECK constraint with syntax and example.
b) List and explain the data types used in oracle.
c) How do create a table from another table ? Explain with syntax and example. (5+5+5)
7. a) Explain UNION and INTERSECT clause used in SQL with example.
b) Explain any 5 aggregate functions with syntax and example in Oracle.



c) Create the table described below : Table name : CLIENT_MASTER.

Column Name	Data type	Size	Attributes
CLIENT_NO	Varchar2	6	Primary key/first letter must start with C, NOT NULL
NAME	Varchar2	20	
ADDRESS1	Varchar2	30	
ADDRESS2	Varchar2	30	
CITY	Varchar2	15	
PINCODE	Number	8	NOT NULL
STATE	Varchar2	15	NOT NULL
BALDUE	Number	10, 2	NOT NULL

Consider the table CLIENT_MASTER given, write the following queries :

- Find out the names of the clients with CLIENT_NO='C00001'.
- Change the city of CLIENT_NO 'C00005' to 'Bangalore'.
- Retrieve the list of names, city and state of all the clients. (5+5+5)

Unit – IV

- Explain the structure of PL/SQL program with example.
 - What is a cursor ? Explain explicit cursor with an example.
 - Explain the syntax and usage of the following with an example :
 - While loop
 - if..end if(5+5+5)
- With syntax and example explain how a package is created and executed in PL/SQL.
 - Write a note on error handling in PL/SQL.
 - With syntax and example explain how a function is created and executed in PL/SQL. (5+5+5)